

Publications

Total publications: 51 / refereed: 35 / first-author: 5 / selected significant contributions: 7 / collaboration papers: 39 / total citations: 6,800 / h-index: 20 (Last updated: 2026-09-13)

First-author publications (citations: 151)

- ⁵ **Anand, Abhijeet**; Aguilar, J.; Ahlen, S.; Bianchi, D.; Brodzeller, A.; *et al.*, 2025, *The Cosmic Evolution of C IV Absorbers at $1.4 < z < 4.5$: Insights from 100,000 Systems in DESI Quasars*, The Astrophysical Journal, **990**, 151 (arXiv:2504.20299) [7 citations] [Impact Factor: 5.7; Scopus Quartile: Q1]

My role: Constructed the largest catalogue of triply ionized carbon absorbers to date and measured the evolution of their incidence and cosmic mass density over $1.4 < z < 4.5$. Interpreted these trends in the context of the cosmic star-formation history and the evolution of the ultraviolet background.

- ⁴ **Anand, Abhijeet**; Guy, Julien; Bailey, Stephen; Moustakas, John; Aguilar, J.; *et al.*, 2024, *Archetype-based Redshift Estimation for the Dark Energy Spectroscopic Instrument Survey*, The Astronomical Journal, **168**, 124 (arXiv:2405.19288) [45 citations] [Impact Factor: 5.8; Scopus Quartile: Q1]

My role: Developed and implemented a new galaxy redshift-fitting method for DESI that reduced catastrophic failures and improved redshift reliability. The method provides a flexible alternative for estimating redshifts in current and future large spectroscopic surveys.

- ³ **Anand, Abhijeet**; Kauffmann, Guinevere; & Nelson, Dylan, 2022, *Cool circumgalactic gas in galaxy clusters: connecting the DESI legacy imaging survey and SDSS DR16 Mg II absorbers*, Monthly Notices of the Royal Astronomical Society, **513**, 3210 (arXiv:2201.07811) [31 citations] [Impact Factor: 5.2; Scopus Quartile: Q1]

My role: Mapped cool metal-enriched gas around galaxy clusters and compared its distribution with that around field galaxies. This was among the first studies to show that galaxy clusters host a substantial reservoir of cool metal-enriched gas in the intracluster medium.

- ² **Anand, Abhijeet**; Nelson, Dylan; & Kauffmann, Guinevere, 2021, *Characterizing the abundance, properties, and kinematics of the cool circumgalactic medium of galaxies in absorption with SDSS DR16*, Monthly Notices of the Royal Astronomical Society, **504**, 65 (arXiv:2103.15842) [65 citations] [Impact Factor: 5.2; Scopus Quartile: Q1]

My role: constructed a large SDSS DR16 Mg II absorber sample to characterize the abundance, spatial distribution, and kinematics of cool circumgalactic gas around star-forming and passive galaxies. Identified the characteristic scales at which differences driven by stellar processes and halo-scale processes begin to emerge.

- ¹ **Anand, Abhijeet**; Roy, Nirupam; & Gopal-Krishna, 2019, *Search for H I emission from superdisk candidates associated with radio galaxies*, Research in Astronomy and Astrophysics, **19**, 083 (arXiv:1812.06875) [3 citations] [Impact Factor: 2.6; Scopus Quartile: Q2]

My role: Analysed VLA observations and performed the spectral analysis to search for H I 21 cm emission from superdisk radio galaxies. Led the interpretation and writing of the paper as part of my MS thesis at IISc Bangalore.

Selected significant contributions (citations: 99)

- ⁷ Napolitano, Lucas; Myers, Adam D.; Tedeschi, Adam; **Anand, Abhijeet**; Herrera-Alcantar, Hiram K.; *et al.*, 2025, *The Composite Spectrum of QSO Absorption Line Systems in DESI DR2*, ArXiv, arXiv:2512.02992 (arXiv:2512.02992) [1 citations] [Impact Factor: N/A; Scopus Quartile: N/A]
My role: Contributed to absorber selection for the stacking analysis and served as an internal reviewer during the collaboration-wide review.
- ⁶ Lan, Ting-Wen; Prochaska, J. Xavier; Aguilar, J.; Ahlen, S.; **Anand, Abhijeet**; *et al.*, 2025, *The Multi-Phase Circumgalactic Medium of DESI Emission-Line Galaxies at $z = 1.5$* , ArXiv, arXiv:2511.03195 (arXiv:2511.03195) [4 citations] [Impact Factor: N/A; Scopus Quartile: N/A]
My role: Contributed to the column density measurements of C IV absorbers identified in DESI quasar spectra and to the interpretation of the results in the context of previous work.
- ⁵ Wu, Xuanyi; Cai, Z.; Lan, T.-W.; Zou, S.; **Anand, Abhijeet**; *et al.*, 2025, *Tracing the Evolution of the Cool Gas in CGM and IGM Environments through Mg II Absorption from Redshift $z = 0.75$ to $z = 1.65$ Using DESI-Y1 Data*, The Astrophysical Journal, **983**, 186 (arXiv:2407.17809) [16 citations] [Impact Factor: 5.7; Scopus Quartile: Q1]
My role: Contributed to the discussion section and to the interpretation of the results in the context of the circumgalactic medium of emission-line galaxies.
- ⁴ Chang, Yu-Ling; Lan, Ting-Wen; Prochaska, J. Xavier; Napolitano, Lucas; **Anand, Abhijeet**; *et al.*, 2024, *Probing the Impact of Radio-mode Feedback on the Properties of the Cool Circumgalactic Medium*, The Astrophysical Journal, **974**, 191 (arXiv:2405.08314) [9 citations] [Impact Factor: 5.7; Scopus Quartile: Q1]
My role: Contributed to writing and to placing the results in the context of previous studies of gas profiles around luminous red galaxies.
- ³ Galiullin, Ilkham; Rodriguez, Antonio C.; El-Badry, Kareem; Szkody, Paula; **Anand, Abhijeet**; *et al.*, 2024, *Searching for new cataclysmic variables in the Chandra Source Catalog*, Astronomy and Astrophysics, **690**, A374 (arXiv:2408.00078) [7 citations] [Impact Factor: N/A; Scopus Quartile: N/A]
My role: Contributed to cross-matching the Chandra Source Catalog with SDSS and DESI EDR data, leading to the identification of new cataclysmic variables with follow-up from eROSITA and Keck.
- ² Napolitano, Lucas; Pandey, Agnesh; Myers, Adam D.; Lan, Ting-Wen; **Anand, Abhijeet**; *et al.*, 2023, *Detecting and Characterizing Mg II Absorption in DESI Survey Validation Quasar Spectra*, The Astronomical Journal, **166**, 99 (arXiv:2305.20016) [31 citations] [Impact Factor: 5.8; Scopus Quartile: Q1]
My role: Contributed to algorithm development and to the physical interpretation of Mg II absorbers detected in DESI early quasar data.
- ¹ Ayromlou, Mohammadreza; Kauffmann, Guinevere; **Anand, Abhijeet**; & White, Simon D. M., 2023, *The physical origin of galactic conformity: from theory to observation*, Monthly Notices of the Royal Astronomical Society, **519**, 1913 (arXiv:2207.02218) [31 citations] [Impact

Factor: 5.2; Scopus Quartile: Q1]

My role: Contributed to the comparison of conformity signals in blue and red galaxies between L-Galaxies simulations and observations from the Legacy Imaging Surveys and SDSS.

Collaboration papers (citations: 6550)

- ³⁹ McArthur, E.; Millon, M.; Powell, M.; Wechsler, R. H.; Pan, Z.; *et al.* (incl. **A. Anand**), 2026, *Quasars Acting as Strong Lenses Found in DESI DR1*, The Astrophysical Journal, **1006**, 147 (arXiv:2511.02009) [1 citations] [Impact Factor: 5.7; Scopus Quartile: Q1]
- ³⁸ Crespi, Andrea; Percival, Will J.; Krolewski, Alex; Bonici, Marco; Zhang, Hanyu; *et al.* (incl. **A. Anand**), 2026, *Baryon fraction from the BAO amplitude: A consistent approach to parametrizing perturbation growth*, Physical Review D, **114**, 043545 (arXiv:2511.23459) [1 citations] [Impact Factor: 5.3; Scopus Quartile: Q1]
- ³⁷ Saydjari, Andrew K.; Draine, Bruce T.; Brandt, Timothy D.; Schlafly, Edward F.; Dey, Arjun; *et al.* (incl. **A. Anand**), 2026, *Spatially-Resolved Spectra of Diffuse Galactic Light using 10.8 M DESI Sky Fibers*, ArXiv, arXiv:2608.28740 (arXiv:2608.28740) [Impact Factor: N/A; Scopus Quartile: N/A]
- ³⁶ Yin, Gaocheng; Jiang, Linhua; Pan, Zhiwei; Martini, Paul; Guo, Wei-Jian; *et al.* (incl. **A. Anand**), 2026, *Luminosity function of quasars at $1.0 < z < 3.5$ from SDSS and DESI*, ArXiv, arXiv:2608.06000 (arXiv:2608.06000) [Impact Factor: N/A; Scopus Quartile: N/A]
- ³⁵ DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alves, O.; *et al.* (incl. **A. Anand**), 2026, *DESI DR2 Results IV: Alcock-Paczyński Measurements from the Lyman Alpha Forest and Cosmological Constraints*, ArXiv, arXiv:2607.27410 (arXiv:2607.27410) [Impact Factor: N/A; Scopus Quartile: N/A]
- ³⁴ Dey, Biprateep; Newman, Jeffrey A.; Zhang, Tianqing; Aguilar, J.; Ahlen, S.; *et al.* (incl. **A. Anand**), 2026, *Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration*, The Astronomical Journal, **172**, 65 (arXiv:2604.06143) [1 citations] [Impact Factor: 5.8; Scopus Quartile: Q1]
- ³³ Moore, Samuel G.; Cole, Shaun; Wilson, Michael; Norberg, Peder; Moustakas, John; *et al.* (incl. **A. Anand**), 2026, *DESI DR2 Galaxy luminosity functions*, Monthly Notices of the Royal Astronomical Society, **549**, stag866 (arXiv:2511.01803) [6 citations] [Impact Factor: 5.2; Scopus Quartile: Q1]
- ³² Negi, Vibhore; Joshi, Ravi; Vishnupriya, B.; Wu, Xue-Bing; Yesuf, Hassen M.; *et al.* (incl. **A. Anand**), 2026, *DESI as sparse Integral Field Spectrograph I: Spatially resolved chemical enrichment in star-forming galaxies at $z \leq 0.1$* , ArXiv, arXiv:2606.12541 (arXiv:2606.12541) [Impact Factor: N/A; Scopus Quartile: N/A]
- ³¹ DESI Collaboration; Abdul Karim, M.; Adame, A. G.; Aguado, D.; Aguilar, J.; *et al.* (incl. **A. Anand**), 2026, *Data Release 1 of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **171**, 285 (arXiv:2503.14745) [526 citations] [Impact Factor: 5.8; Scopus Quartile: Q1]
- ³⁰ Turner, Wynne; Cuceu, Andrei; Martini, Paul; Aguilar, J.; Ahlen, S.; *et al.* (incl. **A. Anand**), 2026, *Probing the limits of cosmological information from the Lyman- α forest 2-point correlation functions*, Journal of Cosmology and Astroparticle Physics, **2026**, 010 (arXiv:2509.14322) [3 citations] [Impact Factor: 6.4; Scopus Quartile: Q1]
- ²⁹ Zaborowski, E. A.; Taylor, P.; Honscheid, K.; Cuceu, A.; de Mattia, A.; *et al.* (incl. **A. Anand**),

- 2026, *H ₀ without the sound horizon (or supernovae): A 2% measurement in DESI DR1*, Journal of Cosmology and Astroparticle Physics, **2026**, 004 (arXiv:2510.19149) [8 citations] [Impact Factor: 6.4; Scopus Quartile: Q1]
- 28 Karaçaylı, Naim Göksel; Ravoux, Corentin; Martini, Paul; Goff, Jean-Marc Le; Armengaud, Eric; *et al.* (incl. **A. Anand**), 2026, *DESI DR1 Ly α 1D power spectrum: Validation of estimators*, Journal of Cosmology and Astroparticle Physics, **2026**, 048 (arXiv:2509.13593) [5 citations] [Impact Factor: 6.4; Scopus Quartile: Q1]
- 27 Ebina, H.; White, M.; Raichoor, A.; Dey, Arjun; Schlegel, D.; *et al.* (incl. **A. Anand**), 2026, *Clustering analysis of medium-band selected high-redshift galaxies*, Journal of Cosmology and Astroparticle Physics, **2026**, 019 (arXiv:2509.26467) [3 citations] [Impact Factor: 6.4; Scopus Quartile: Q1]
- 26 Hsu, Yuan-Ming; Huang, Xiaosheng; Storfer, Christopher J.; Inchausti, Jose Carlos; Schlegel, David; *et al.* (incl. **A. Anand**), 2026, *A New Way to Discover Strong Gravitational Lenses: Pairwise Spectroscopic Search from DESI DR1*, The Astrophysical Journal Supplement Series, **282**, 41 (arXiv:2509.16033) [2 citations] [Impact Factor: 7.7; Scopus Quartile: Q1]
- 25 Han, Jiwon Jesse; Chiti, Anirudh; Chen, Kai-Feng; Bechtol, Keith; Bellini, Andrea; *et al.* (incl. **A. Anand**), 2026, *A Path to an All-Sky Survey with Roman*, ArXiv, arXiv:2602.21280 (arXiv:2602.21280) [1 citations] [Impact Factor: N/A; Scopus Quartile: N/A]
- 24 Kamalinejad, Farshad; Slepian, Zachary; Krolewski, Alex; Greco, Alessandro; Ortolá Leonard, William; *et al.* (incl. **A. Anand**), 2026, *First Detection of the Baryon Acoustic Oscillation (BAO) Feature in the 3-Point Correlation Function of DESI DR1 Luminous Red Galaxies*, ArXiv, arXiv:2602.16134 (arXiv:2602.16134) [1 citations] [Impact Factor: N/A; Scopus Quartile: N/A]
- 23 Lange, Johannes U.; Wells, Alexandra; Hearin, Andrew; Beltz-Mohrmann, Gillian; Leauthaud, Alexie; *et al.* (incl. **A. Anand**), 2025, *Cosmological Constraints from Full-Scale Clustering and Galaxy-Galaxy Lensing with DESI DR1*, ArXiv, arXiv:2512.15962 (arXiv:2512.15962) [6 citations] [Impact Factor: N/A; Scopus Quartile: N/A]
- 22 Chang, Yu-Ling; Lan, Ting-Wen; Prochaska, J. Xavier; Siudek, Malgorzata; Aguilar, J.; *et al.* (incl. **A. Anand**), 2025, *Tracing the Cosmic Evolution of the Cool Circumgalactic Medium of Luminous Red Galaxies with DESI Year 1 Data*, ArXiv, arXiv:2512.03845 (arXiv:2512.03845) [4 citations] [Impact Factor: N/A; Scopus Quartile: N/A]
- 21 Herrera-Alcantar, Hiram K.; Armengaud, Eric; Yèche, Christophe; Gordon, Calum; Casas, Laura; *et al.* (incl. **A. Anand**), 2025, *The Lyman- α forest from LBGs: First 3D correlation measurement with DESI and prospects for cosmology*, Journal of Cosmology and Astroparticle Physics, **2025**, 053 (arXiv:2507.21852) [4 citations] [Impact Factor: 6.4; Scopus Quartile: Q1]
- 20 Blanco, Diana; Leauthaud, Alexie; Ulf Lange, Johannes; Wright, Angus H.; Hildebrandt, Hendrik; *et al.* (incl. **A. Anand**), 2025, *The Power of DESI for Photometric Redshift Calibration: A Case Study with KiDS-1000*, ArXiv, arXiv:2512.15964 (arXiv:2512.15964) [3 citations] [Impact Factor: N/A; Scopus Quartile: N/A]
- 19 Uzsoy, Ana Sofia M.; Dey, Arjun; Raichoor, Anand; Finkbeiner, Douglas P.; Ramakrishnan, Vandana; *et al.* (incl. **A. Anand**), 2025, *Effect of local environment on Ly α line profile in DESI/ODIN LAEs*, ArXiv, arXiv:2511.17498 (arXiv:2511.17498) [3 citations] [Impact Factor: N/A; Scopus Quartile: N/A]
- 18 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Alam, S.; Allen, L.; *et al.* (incl. **A. Anand**), 2025, *DESI*

- DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints*, Physical Review D, **112**, 083515 (arXiv:2503.14738) [1366 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]
- 17 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Allende Prieto, C.; Alves, O.; *et al.* (incl. **A. Anand**), 2025, *DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest*, Physical Review D, **112**, 083514 (arXiv:2503.14739) [252 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]
 - 16 Brodzeller, A.; Wolfson, M.; Santos, D. M.; Ho, M.; Tan, T.; *et al.* (incl. **A. Anand**), 2025, *Construction of the damped Ly α absorber catalog for DESI DR2 Ly α BAO*, Physical Review D, **112**, 083510 (arXiv:2503.14740) [24 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]
 - 15 Rashkovetskyi, Michael; Eisenstein, Daniel J.; Aguilar, Jessica Nicole; Ahlen, Steven; **Anand, Abhijeet**; *et al.*, 2025, *Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect*, The Open Journal of Astrophysics, **8**, 46033 (arXiv:2508.20904) [**Impact Factor: 2.4; Scopus Quartile: Q1**]
 - 14 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 008 (arXiv:2411.12021) [179 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]
 - 13 Treiber, Helena; Amon, Alexandra; Wechsler, Risa H.; Manwadkar, Viraj; Myles, Justin; *et al.* (incl. **A. Anand**), 2025, *Dwarf galaxy halo masses from spectroscopic and photometric lensing in DESI and DES*, ArXiv, arXiv:2509.20434 (arXiv:2509.20434) [2 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]
 - 12 Cuceu, Andrei; Herrera-Alcantar, Hiram K.; Gordon, Calum; Ramírez-Pérez, César; Armengaud, E.; *et al.* (incl. **A. Anand**), 2025, *DESI DR1 Ly α forest: 3D full-shape analysis and cosmological constraints*, ArXiv, arXiv:2509.15308 (arXiv:2509.15308) [20 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]
 - 11 Medina, Gustavo E.; Li, Ting S.; Eadie, Gwendolyn M.; Riley, Alexander H.; Valluri, Monica; *et al.* (incl. **A. Anand**), 2025, *The mass of the Milky Way from outer halo stars measured by DESI DR1*, ArXiv, arXiv:2508.19351 (arXiv:2508.19351) [4 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]
 - 10 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements*, Journal of Cosmology and Astroparticle Physics, **2025**, 028 (arXiv:2411.12022) [333 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]
 - 9 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 II: sample definitions, characteristics, and two-point clustering statistics*, Journal of Cosmology and Astroparticle Physics, **2025**, 017 (arXiv:2411.12020) [129 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]
 - 8 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 III: baryon acoustic oscillations from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 012 (arXiv:2404.03000) [546 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]
 - 7 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations*,

- Journal of Cosmology and Astroparticle Physics, **2025**, 021 ([arXiv:2404.03002](#)) [[1774 citations](#)] [**Impact Factor: 6.4; Scopus Quartile: Q1**]
- ⁶ Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest*, Journal of Cosmology and Astroparticle Physics, **2025**, 124 ([arXiv:2404.03001](#)) [[381 citations](#)] [**Impact Factor: 6.4; Scopus Quartile: Q1**]
 - ⁵ Ross, A. J.; Aguilar, J.; Ahlen, S.; Alam, S.; **Anand, Abhijeet**; *et al.*, 2025, *The construction of large-scale structure catalogs for the Dark Energy Spectroscopic Instrument*, Journal of Cosmology and Astroparticle Physics, **2025**, 125 ([arXiv:2405.16593](#)) [[58 citations](#)] [**Impact Factor: 6.4; Scopus Quartile: Q1**]
 - ⁴ Scholte, Dirk; Saintonge, Amélie; Moustakas, John; Catinella, Barbara; Zou, Hu; *et al.* (incl. **A. Anand**), 2024, *The atomic gas sequence and mass-metallicity relation from dwarfs to massive galaxies*, Monthly Notices of the Royal Astronomical Society, **535**, 2341 ([arXiv:2408.03996](#)) [[29 citations](#)] [**Impact Factor: 5.2; Scopus Quartile: Q1**]
 - ³ DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; *et al.* (incl. **A. Anand**), 2024, *The Early Data Release of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **168**, 58 ([arXiv:2306.06308](#)) [[565 citations](#)] [**Impact Factor: 5.8; Scopus Quartile: Q1**]
 - ² DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; *et al.* (incl. **A. Anand**), 2024, *Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **167**, 62 ([arXiv:2306.06307](#)) [[300 citations](#)] [**Impact Factor: 5.8; Scopus Quartile: Q1**]
 - ¹ Han, Jiwon Jesse; Dey, Arjun; Price-Whelan, Adrian M.; Najita, Joan; Schlafly, Edward F.; *et al.* (incl. **A. Anand**), 2023, *NANCY: Next-generation All-sky Near-infrared Community survey*, ArXiv, [arXiv:2306.11784](#) ([arXiv:2306.11784](#)) [[10 citations](#)] [**Impact Factor: N/A; Scopus Quartile: N/A**]